

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010

Data Analysis Report

Project ID: 25-26J-133

Project Title: AI-Driven Innovations to Enhance Tea Production and Quality in Sri Lanka's
Mid-Country

1. Introduction

1.1. Background:

Sri Lanka's plantation economy remains strongly supported by the tea sector, which contributes substantially to export revenue and rural livelihoods, particularly through smallholder producers. However, decision-making across the tea value chain is frequently constrained by manual inspection practices, delayed reporting, and limited access to field-level analytics. These limitations reduce consistency in quality assessment, weaken early intervention in crop risks, and restrict market responsiveness.

Four operational gaps are commonly observed. First, freshness grading and tea-type identification are often performed manually, resulting in variability across inspectors and locations. Second, yield planning is affected by uncertainty in weather, field conditions, and management practices, limiting the ability to anticipate crop output and optimize resource allocation. Third, demand estimation and auction planning typically rely on historical trends and expert judgment, which can be insufficient under market volatility. Fourth, disease identification and treatment guidance are often delayed due to limited expert availability, leading to avoidable yield losses and quality degradation.

To address these issues, a mobile-based, data-driven decision support application is being developed as a student-led data science project, with a focus on accessibility for smallholder communities. The system is designed to integrate (i) automated freshness grading and tea-type classification, (ii) predictive yield analytics, (iii) market demand forecasting for smarter auction planning, and (iv) AI-based tea leaf disease and pest detection with treatment guidance. By enabling rapid field-level assessments and actionable recommendations, the proposed solution aims to improve consistency, efficiency, and scalability of tea-sector decision-making in Sri Lanka.

1.2. Research Problem:

The Sri Lankan tea industry, though vital to the economy, struggles with productivity, quality, and sustainability. Farmers face inconsistent plucking, quality loss, and undetected disease and pest outbreaks, while factories and auction stakeholders

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depend on manual grading, limited forecasting, and inefficient planning, leading to reduced competitiveness.

1.3. Objectives:

- 1) To develop an automated tea leaf freshness grading and tea type classification module using computer vision techniques, reducing subjectivity and variability associated with manual inspection.
- 2) To implement tea yield analytics and digitization module that converts handwritten logbooks into structured datasets and models environmental, management, and historical yield factors to deliver accurate forecasts and support data-driven estate planning.
- 3) To forecast Sri Lanka's tea market demand and support smart auction planning by developing a weekly, elevation-aware decision-support framework that integrates time-series forecasting, machine learning, and auction-behavior simulation to capture long-term market trends, short-term demand shocks, and seasonal variability.
- 4) To design an AI-based tea leaf disease, pest, invasive species, and seasonal outbreak prediction module capable of identifying locally specific major leaf and pest diseases from images, detecting invasive plants, and forecasting seasonal disease risk patterns, while providing actionable, expert-aligned treatment and management recommendations.
- 5) To integrate all analytical components into a mobile application that enables real-time, field-level access for smallholder farmers with minimal technical expertise.

It Number	Objective	Objective number
IT22154576	Automated Tea Freshness Grading and	1
IT2222268	Predictive Analytics for Tea Crop Yield Management	2
IT 22247018	Tea Market Demand Forecasting and Smart Auction Planning	3
IT22142146	AI-Based Leaf Disease, Pest & Invasive Species Detection and Treatment Guidance	4

2. Data Exploration

2.1. Data Collection

To build a real-world dataset, we visited smallholder tea estates and collected freshly plucked leaves directly from the field. For each sample, the leaves were spread in a single layer on a **clean, light-coloured (white) cloth** to create a consistent, low-noise background. Images were captured **outdoors under daylight** using our dedicated data-

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collection mobile application, keeping the camera angle and distance as consistent as possible across sessions. This setup reduces background variation and helps the model focus on leaf colour, texture, and shape—while still preserving natural lighting changes that occur in practical field conditions.

In addition, we visited several tea estates to gain field-level knowledge and capture real plantation images of major locally relevant tea leaf diseases, pests, and invasive plants. These field photographs were further supplemented with Kaggle datasets and curated agricultural references to improve dataset diversity.

2.2. Dataset Description:

Data source	Description	Resource	Size	Key attributes
Self-collected Tea Leaf Image Dataset (Smallholder Estates)	Field-collected image dataset of freshly harvested tea leaves captured under natural daylight conditions for quality and freshness analysis	Dataset Custom mobile data-collection application; locally stored image dataset	275 original images, augmented to 1,000 samples for model training	RGB images, daylight illumination, uniform background, leaf colour, texture, shape, freshness labels
Tea types dataset	Curated image dataset representing different processed tea types used for tea type classification	Publicly available datasets and curated online sources; locally consolidated dataset	Pending	RGB images, tea type labels (e.g., black, green, white), colour variation, texture patterns, processed leaf characteristics

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Self-collected Tea Yield Logbook Images Dataset (Mid-Country Estates)	Self-collected images of manually written pruning-cycle yield log books from two mid-country estates, captured for OCR-based digitization and scalable to other estates using the same standard template.	Self-collected estate log book images (local storage)	175 log book images from 2 estates in the mid country	Division/field IDs, dates, monthly and cycle yields, bush age, rainfall, wet days, pruning year/style, fertilizer and dolomite entries, Soil pH, remarks
Mid-Country Estate-Level Monthly Yield Dataset (Field-wise, 10 Years)	Field-wise historical monthly yield data with environmental and management factors	field-wise-yield-dataset.xlsx	5,821 rows and 22 columns across 10 years	Division, Field_ID, Year, Month, Planting_Year, bush age, pruning info, rainfall, wet days, fertilizer details, plucking rounds, soil pH, Yiled total
Tea Auction Data transaction Dataset 2023-2025	Colombo Tea Auction transaction records	DATA TABLE-2023-2025 SALE 50.xlsx (Sheet: DATA)	1.5M rows, 26 cols across 3 years	YEAR, SALE NO, GRADE, QTY, PRICE, ELE1, BROKER, FACTORY, BUYER, MFREF
TeaLeaves Dataset	Image dataset	Collected from local tea estates, and supplemented	~6,000+ annotated	Disease class label,

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	containing healthy tea leaves and major tea leaf diseases and pests. Includes common field level conditions and symptom variations. - Blister Blight - Brown Blight - Grey Blight - Red Rust - Healthy - Helopeltis	with Kaggle datasets and curated academic references.	images across 6 classes (Healthy, Blister Blight, Brown Blight, Grey Blight, Red Rust, Helopeltis)	RGB images - Color images used for visual pattern recognition, leaf texture - Used by CNN to identify disease patterns, color variation, visible symptoms, background variation
InvasiveSpecies Dataset	Image dataset containing invasive plant species affecting tea plantations. - Lantana camara - Tridax procumbens	Field photographed from tea growing regions, supplemented with Kaggle invasive species datasets and curated online/agricultural reference sources.	~1,500+ annotated images across 2 invasive species classes (Lantana camara, Tridax procumbens)	Species label, RGB image, leaf shape, texture patterns, color variations, growth characteristic, background conditions
Tea Season Disease Dataset	Structured dataset	tea_season_disease_dataset.xlsx	5,000+ rows with	Month, Season, Region,

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	containing seasonal and environmental conditions linked with major tea leaf diseases, used for predicting disease risk levels across different months and regions in Sri Lanka.	(locally prepared dataset based on Sri Lankan seasonal patterns and estate level observations)	8 key columns	Temperature, Humidity, Rainfall, Disease_Label, Risk_Level
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2.3. Suitability Analysis

2.3.1. Relevance to Individual Research Objectives:

(Explain how well each dataset aligns with your research problem and objectives)

	1	2	3	4
Data source 1: Tea Freshness dataset	✓			
Data source 2: Self-collected Tea Yield Logbook Images Dataset (Mid-Country Estates)		✓		
Data source 3: Mid-Country Estate-Level Monthly Yield Dataset (Field-wise, 10 Years)		✓		
Data source 4: Auction			✓	

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transaction data (2023–2025)				
Data source 5: TeaLeaves Dataset				✓
Data source 6: InvasiveSpecies Dataset				✓
Data source 7: Tea Season Disease Dataset				✓

3. Methodology

3.1. Data Preprocessing:

(Mention data transformation techniques done in each dataset for each objective.)

Ex:

Data Cleaning, Data Normalization, Data Standardization, Data Encoding (e.g., One-Hot Encoding, Label Encoding), Handling Missing Data (e.g., Imputation or Removal), Data Aggregation, Feature Engineering, Outlier Detection and Handling, Data Scaling, Data Discretization, Dimensionality Reduction (e.g., PCA), Date/Time Transformation, Data Integration (Merging or Joining), Data Mapping, Data Type Conversion.

Objective 1

	Transformation technique
Data source 1	- Image Resizing - Pixel Normalization - Colour Space Conversion - Grayscale Conversion - Data Augmentation - Background Masking

Objective 3

Data source	Transformation technique
Data source 1: Auction Transaction Dataset (2023–2025 Excel)	Schema Harmonization (standardize column names across files) Type Casting (PRICE, QTY → numeric using <code>to_numeric(errors="coerce")</code>) Missing / Invalid Handling (drop/flag rows with missing PRICE or QTY) Text Cleaning (trim spaces, unify GRADE labels, fix case/typos) Elevation Validation (keep only ELE1 ∈ {Low, Medium, High})

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	• Duplicate Removal (remove repeated rows after merging files) • Outlier Control (flag/cap extreme PRICE and very low QTY to reduce distortion) • Weekly Aggregation (group by YEAR + SALE NO + GRADE (+ ELE1); compute $\text{avg_price} = \frac{\sum(\text{PRICE} \times \text{QTY})}{\sum(\text{QTY})}$ and $\text{total_qty} = \sum(\text{QTY})$) • Week-Date Mapping (convert SALE NO to a continuous weekly date index; handle year boundaries)
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Objective 4: AI-Based Leaf Disease, Pest and Treatment Guidance

	Transformation technique
Data source 5: TeaLeaves Dataset	• Image Resizing (224×224 for CNN input) • Pixel Normalization (0–1 scaling) • Label Encoding (6 disease/pest classes) • Data Augmentation (rotation, flipping, brightness adjustment) • Noise Reduction (blur filtering for unclear samples) • Background Variation Handling (field-level robustness)
Data source 6: InvasiveSpecies Dataset	• Image Resizing (CNN compatible size) • Normalization and Standardization • Class Label Encoding (2 invasive species categories) • Data Augmentation (zoom, rotation, contrast adjustment) • Occlusion Handling (training with complex plantation backgrounds)
Data source 7: Seasonal Risk Prediction Dataset	• Missing Value Handling (imputation for climate gaps) • Feature Encoding (Season, Region → Label/One-Hot Encoding) • Data Scaling (Temperature, Humidity, Rainfall normalization) • Outlier Detection (extreme rainfall/temperature removal) • Train–Test Splitting (balanced disease-risk distribution)

3.2. Scalability

Data source 1:

The self-collected Tea Leaf Image Dataset currently contains 275 original images augmented to 1,000 samples. The data collection infrastructure is designed for longitudinal scalability

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through a batch management system with unique batch identifiers (BATCH_YYYYMMDD_HHMMSS), sequential image indexing for traceability, and JSON sidecar metadata files that enable automated ingestion into database systems. Automated quality gates maintain dataset consistency as collection scales across multiple operators, with rejection logging identifying systematic capture issues for operator training. The TeaVision mobile application supports configurable capture resolutions (2 MP to 12 MP+) accommodating diverse device capabilities, while asynchronous processing with coroutine-based concurrency maintains UI responsiveness. The standardized input preprocessing pipeline is compatible with pre-trained CNN architectures (VGG, ResNet, MobileNet) enabling transfer learning, and batch-organized data supports chronological train/validation splits for incremental learning as the dataset grows.

Data source 2:

The Pruning Cycle Yield Logbook dataset currently includes 175 images across two mid-country estates. In this dataset there are two distinct pruning-cycle templates, each corresponding to a different plantation, and within each plantation the same template is reused for all estates. Together these two templates cover approximately 35 estates, so once the digitization pipeline is configured for each template, it can scale by simply ingesting additional estate yield books without changing the extraction logic.

Data source 3:

The monthly yield forecasting dataset consists of 5,821 field-wise records with 22 variables from a single mid-country estate over 2012–2025, providing enough historical coverage and management variation to train and validate estate-level monthly yield prediction models. Each new year's data can be appended as additional rows to this dataset as it becomes available.

Data source 4:

The Auction transaction dataset is large 1.5M rows across 3 years, which is sufficient for robust weekly aggregation and time-series training. Weekly aggregation reduces noise and compresses the dataset into manageable time-series sequences per grade/elevation. The system is scalable by adding new yearly auction files and retraining models periodically.

Data source 5:

The **TeaLeaves Dataset**, which contains more than 6,000 labeled images across six main tea leaf disease and pest categories, gives a good amount of data to train CNN and object detection models effectively. Since the images come from local tea estates and Kaggle, the dataset includes different lighting conditions, symptom patterns, and backgrounds, making it useful for real field applications.

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Data source 6:

The **InvasiveSpecies Dataset**, with over 1,500 images of Lantana camara and Tridax procumbens, provides a strong starting point for detecting invasive species. It can be improved using data augmentation and by collecting more invasive species images from Sri Lankan plantations. Overall, both TeaLeaves Dataset and InvasiveSpecies Datasets can be extended later to include more diseases, pests, and invasive plant species, which supports future improvements and real-world use of the system.

Data source 7:

The **Tea Season Disease Dataset**, with around 5,000 records, provides a solid base for seasonal tea disease risk prediction using machine learning. It includes key factors such as month, season, region, temperature, humidity, rainfall, and disease labels, helping the models learn climate-related disease patterns. The dataset is scalable because more years of weather data, additional regions, and new disease categories can be added easily in the same tabular structure, allowing continuous improvement through future expansion and retraining.

3.3. Feature extraction

Objective 1: Tea Leaf Freshness Grading

A total of 25 hand-crafted features were extracted from each leaf image across four categories:

Colour Features (6): Mean Hue, Mean Saturation, Mean Value (HSV colour space over leaf-masked pixels), Greenness Index (composite metric: $(1 - |\text{meanHue} - 120| / 60) \times \text{meanSaturation}$), Brownness Index (oxidation indicator for $\text{hue} \in [20^\circ, 60^\circ]$), and Colour Uniformity ($1 / (1 + \text{hueVariance} / 1000)$).

Texture Features (2): Blur Score (Laplacian variance $\times 10$ on 400px-width scaled image) and Background Variance (standard deviation of luminance in peripheral 15% edge regions).

Morphological Features (6): Leaf Area (pixels and cm^2), Leaf Width (cm), Leaf Height (cm), Leaf Coverage (%), and Aspect Ratio — derived from binary segmentation masks with optional QR-code calibration for real-world measurements.

Quality Metrics (4): Resolution (MP), Brightness (mean luminance), Glare Percentage, and Segmentation Confidence (0.9 for ML-based, 0.7 for colour-based fallback).

Segmentation uses a dual approach: a TensorFlow Lite model (256 $\times$ 256 input, softmax threshold 0.5) as the primary method, with a colour-based fallback using adaptive brightness thresholds and greenish-tint validation for devices without ML acceleration.

Objective 2: Predictive Analytics for Tea Crop Yield Management

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Objective 3: Tea Market Demand Forecasting and Smart Auction Planning

A total of 30 engineered time-series and market-intelligence features were extracted for each Grade × Elevation × Week record to forecast demand signals (price pressure + buyer activity) and support smart auction planning. Features were derived across five categories:

Market Price Dynamics : Lagged and trend-based price indicators were computed to capture short-term and seasonal market movement: Price($t-1$), Price($t-2$), Price($t-4$), 4-week Rolling Mean, 8-week Rolling Mean, 4-week Rolling Std (volatility), Week-over-Week Price Change (%), and Momentum Index (*rolling mean of WoW change over last 3 weeks*).

Supply & Volume Pressure Features: Auction supply signals were modelled using quantity-based indicators: Offered Quantity($t-1$), ($t-2$), ($t-4$), 4-week Rolling Mean Quantity, Quantity Shock Flag (*1 if quantity > mean + 1.5×std*), Scarcity Index ($1 / (1 + \text{normalized quantity})$) and Grade-Mix Share (*grade quantity / total weekly quantity within the elevation category*).

Buyer Demand & Competition Features: Demand strength was captured using buyer-side proxies: Active Buyer Count, Top-3 Buyer Share, Buyer Concentration (HHI), Competition Ratio (*buyers / lots*), Repeat-Buyer Rate (*buyers appearing in both current and previous week*), and Bid Intensity Proxy (*normalized price increase under limited supply conditions*).

Demand Target Construction (Forecast Label) : For modelling demand, the target variable was defined as a Demand Pressure Score, computed using a weighted combination of price uplift, buyer participation, and supply scarcity (higher prices with lower offered quantities and higher buyer competition imply stronger demand). This demand signal is then used to recommend **optimal offering** quantities, grade prioritization, and reserve price guidance for smart auction planning.

Objective 4: AI-Based Leaf Disease, Pest and Treatment Guidance

Image-Based Feature Extraction: For tea leaf disease, pest, and invasive plant identification, feature extraction was performed using CNN models. It automatically learns discriminative visual characteristics from input images.

- **Colour Variations:** rust patches, browning, yellowing, and abnormal discoloration caused by infections.
- **Symptom and Spot Patterns:** blight lesions, fungal marks, and irregular disease spread across leaf surfaces.
- **Shape and Structural Features:** leaf deformation, holes, and chewing marks linked to pest activity.

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- Invasive Species Characteristics: distinctive leaf shape, vein structure, and plant morphology for species such as Lantana camara and Tridax procumbens.

Feature Preparation and Enhancement: To improve robustness under real field conditions,

- Resizing and pixel normalization
- Data augmentation (rotation, flipping, brightness adjustments)
- Noise and background variation reduction

Seasonal Disease Feature Extraction: For seasonal disease prediction dataset, structured feature extraction was performed using key climatic, temporal variables such as,

- Month and seasonal indicators
- Regional conditions
- Temperature, humidity, and rainfall patterns
- Disease risk labels

4. Modelling and Results

The results section should present the key findings derived from analyzing the data. It should answer the research questions or hypotheses posed at the beginning of the study, based on the methods and techniques applied during the analysis.

4.1. Key Insights:

Data Quality Analysis (Objective 1: Tea Leaf Freshness Grading):

Analysis of the quality assessment pipeline across collected samples showed that the overall quality pass rate was 78.4%, with individual metrics as follows: Resolution pass rate 98.5%, Blur Score 85.2%, Brightness 91.3%, Glare 94.7%, and Background 88.9%. Multiple simultaneous failures were rare, indicating the quality gates operate independently and effectively.

Quality Metric	Pass Rate	Primary Failure Mode
Resolution	98.5%	Operator selection of low-resolution mode
Blur Score	85.2%	Handheld motion during capture
Brightness	91.3%	Shaded capture locations
Glare	94.7%	Direct sunlight on leaf surface

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Background	88.9%	Insufficient background cloth coverage
Overall Pass	78.4%	Multiple simultaneous failures rare

Data Quality Analysis (Objective 2: Predictive Analytics for Tea Crop Yield Management):

Data Quality Analysis (Objective 3: Tea Market Demand Forecasting & Smart Auction Planning)

An assessment of the auction + external feature engineering pipeline showed that most records were usable for weekly demand forecasting, with issues mainly coming from missing price/quantity entries, inconsistent grade/elevation labeling, and macro-data alignment gaps. Importantly, multi-metric failures were uncommon, suggesting that the quality gates capture *different* independent failure modes (good for reliable filtering rather than over-filtering).

Quality Metric	Pass Rate	Primary Failure Mode
Year Validity (2000–2100)	100.0%	Not observed (all YEAR values valid)
Sale/Week Validity (1–60)	100.0%	Not observed (all SALE NO values valid)
Grade Presence	100.0%	Not observed (GRADE always present)
Elevation Validity (Low/Medium/High)	100.0%	Not observed (ELE1 consistently labeled)
Quantity Numeric + Positive	100.0%	Not observed (QTY clean and >0)
Price Numeric + Positive	100.0%	Not observed (PRICE clean and >0)
Price Plausibility Gate (100–10,000 LKR)	98.98%	Rare extreme high prices (special lots / outliers / entry scale issues)
Quantity Plausibility Gate (≥ 1 unit)	99.99%	Very small QTY likely representing sample lots / unit inconsistencies
Weekly Aggregation Validity	100.0%	Not observed (weekly calculations stable)

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Weekly Avg-Price Plausibility (100–10,000 LKR)	91.67%	Weekly averages inflated when very high price combines with low total_qty
Overall Pass (Core + Plausibility gates)	98.98%	Dominated by the price outlier cases

Data Quality Analysis (Objective 4: AI-Based Leaf Disease, Pest and Treatment Guidance)

Most disease, pest, invasive species, and seasonal prediction samples were suitable for CNN based modelling. The main challenges came from lighting variations, similar symptoms between some diseases, and occasional background noise in field images. Overall, quality checks ensured reliable data for training.

Quality Metric	Pass Rate	Primary Issue
Label Availability (6 Classes)	100%	All images correctly labeled
Resolution Validity ($\geq 224 \times 224$)	97.6%	Few low-resolution samples
Blur Detection	89.4%	Motion blur during capture
Lighting Consistency	90.8%	Overexposure in direct sunlight
Background Noise Control	88.7%	Complex field backgrounds
Overall Pass Rate	91.2%	Mainly affected by blur + background variation

Tea Leaf Disease & Pest Detection Results: A CNN-based classifier trained on the TeaLeaves dataset (~6,000 images across 6 classes) achieved strong performance in identifying healthy leaves and major Sri Lankan tea diseases and pests.

Model Metric	Result	Key Observation
Training Accuracy	~95%	Learned disease-specific visual patterns effectively
Validation Accuracy	~88–90%	Generalized well under field-level variations
Most Confused Classes	Blister vs Brown Blight	Similar lesion patterns caused minor overlap

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Invasive Species Detection Quality: The InvasiveSpecies dataset (~1,500 images of Lantana camara and Tridax procumbens) provided a solid foundation for weed detection with minor errors.

Quality Metric	Pass Rate	Primary Issue
Species Label Presence	100%	All samples labeled correctly
Shape Feature Visibility	91.8%	Partial occlusion in dense weeds
Overall Pass Rate	92.0%	Mostly affected by background complexity

4.2. Challenges Faced During Data Analysis:

The analysis of our datasets presented several challenges, primarily during the data gathering and preprocessing phases. The key challenges encountered include:

1. Data Collection Challenges

- Freshness grade labelling needed domain expert judgment, which introduces subjectivity. Different assessors may give different grades to borderline samples. To reduce this, all 275 original labels were manually audited for consistency.
- Early tea leaf collection was biased toward fresh samples because field visits coincided with recent plucking. Fresh leaves were easy to find, while aged or damaged samples had to be collected deliberately.
- Although public datasets were available, collecting data that matches Sri Lankan tea estate conditions was difficult. Field data collection required consistent lighting, camera angles, and backgrounds.
- In some cases, certain disease, pest, or invasive species categories had fewer samples, creating class imbalance.
- Invasive species images also required careful collection because plants were often mixed with surrounding weeds.
- Auction data came from several Excel files (2023–2025). Even though the columns were mostly the same, small formatting differences (extra spaces, numbers stored as text,

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naming differences) had to be cleaned before merging, and that can hide small inconsistencies.

- Real demand data (actual bids, buyer competition, order books, broker negotiations) were not available. So demand had to be estimated using indirect signals like price changes and supply shortages, which is less accurate than having true buyer-side data.
- Some tea grades do not appear every week in each elevation (Low/Medium/High). This creates short or broken time series for those grades, making forecasting harder and forcing us to remove grades with too few data points.
- The dataset uses SALE NO instead of normal calendar weeks. Converting SALE NO into dates requires assumptions and careful handling across year changes, and small timing mismatches can affect alignment with external data.

2. Data Quality and Variability

- Different smartphone cameras reproduce colors differently and use different white balance and exposure settings. This affects how well the model works across devices, especially for colour-based features like greenness and brownness indices.
- Outdoor lighting changes (shadows from trees, cloud cover, time-of-day shifts) cause brightness to vary by about ± 30 units, so preprocessing normalization is needed before extracting colour features.
- The datasets contained variations in image quality, background noise, and symptom appearance. Tea leaf diseases and pest damage often look similar, such as Blister Blight and Brown Blight share similar visual symptoms, which made classification more challenging.
- Auction and yield datasets also required careful checking due to inconsistent formatting across sources.

3. Preprocessing and Data Cleaning Issues

Overlapping leaves, curled edges, and shadows create unclear boundaries during segmentation, which affects the accuracy of leaf measurements like area, width, and height.

Session-based data splitting is needed to prevent data leakage. Images from the same capture session share lighting, angles, and leaf arrangement, so randomly splitting across sessions would give misleadingly high performance numbers.

- Several preprocessing steps were needed, such as image resizing, normalization, augmentation, and label verification.
- Background complexity and early-stage infections sometimes reduced symptom visibility, affecting feature learning.
- Seasonal prediction data also required consistency checks across climatic variables and risk-level distributions.

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- For yield analytics, handwritten logbooks required OCR-based digitization, where unclear handwriting and missing entries affected extraction accuracy.

4. Handling Large-Scale Data

- The auction transaction dataset included over 1.5 million records, which increased processing time and required aggregation into weekly trends to make forecasting manageable.

5. Computational and Training Limitations

The current dataset of only 275 original images (augmented to 1,000) is relatively small for deep learning, which limits how well the model can generalize to new, unseen data. Continued data collection across more estates and seasons is planned to address this.

- Training CNN based models for disease and invasive species detection required significant computational resources.
- Limited hardware availability increased training time and restricted the number of experiments and hyperparameter tuning that could be performed.

5. References

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